

QuantOS

Product Requirements Document (PRD) — Production Specification

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1. Executive Summary & Vision

QuantOS is an offline-first, local-data-driven desktop operating system engineered for mastering Quantitative Research and Quantitative Finance. Unlike generic note-taking apps, standard task managers, or habit trackers, QuantOS provides an opinionated, graph-based curriculum engine. It actively computes daily learning schedules, tracks prerequisite completion across a strict Directed Acyclic Graph (DAG), enforces mathematical spaced repetition (SM-2 variant), and calculates real-time learning velocity against target completion dates.

THE 10 DAILY CORE QUESTIONS QUANTOS SOLVES DETERMINISTICALLY

- What should I study today?** Computed daily by DAG availability, SM-2 decay, and velocity targets.
- What topics are unlocked?** Evaluated in real time via DAG node dependency resolution.
- What prerequisites am I still missing?** Mapped directly per node in the visual dependency graph.
- Which resources should I use?** Curated, rated, and categorized per curriculum node.
- Which projects should I complete?** Practical programming assignments mapped directly to nodes.
- What should I revise today?** Automated SM-2 revision queue ordered by mathematical decay curves.
- How many days behind schedule am I?** Calculated via rolling 14-day velocity vs. remaining estimated hours.
- How much of the roadmap is completed?** Weighted node completion percentage across all 81 topics.
- How much time have I invested?** Measured via active precision study timers and verified logs.
- What should I study tomorrow?** Forecasted 24-hour lookahead queue based on active throughput.

2. Product Goals & Success Metrics

Metric ID	KPI Description	Target Threshold	Measurement Method
KPI-01	Daily Active Engagement	≥ 5 active study days / week	Local database activity log audit
KPI-02	Plan Execution Rate	≥ 85% tasks completed as scheduled	Ratio of completed vs scheduled daily queue tasks
KPI-03	Revision Compliance	≥ 90% SM-2 reviews cleared on due date	Spaced repetition queue lag tracking
KPI-04	Schedule Pace Accuracy	Deviation ≤ 15% from forecasted end date	Rolling 14-day velocity vs remaining hours estimate
KPI-05	UI Performance & Startup	< 300 ms cold start; < 16 ms render frame	Local application performance profiling

3. Target Persona & Workflows

Primary Target Persona: The Quant Researcher Candidate ("The Candidate"). A student or practitioner in Computer Science, Mathematics, Physics, or Engineering aiming for quantitative investment firms, prop desks, and hedge funds.

Core Workflow: Launch QuantOS → Check Daily Focus Engine → Execute SM-2 Mathematical Revisions → Study Unlocked DAG Topic Node → Run Precision Session Timer → Log Daily Reflection & Qualitative Review → System Recalculates DAG & Velocity Forecast.

4. Curriculum Architecture & Dependency Graph

The curriculum consists of **11 Phases** containing **81 Nodes** totaling **1,970 Estimated Study Hours** structured as a strict Directed Acyclic Graph (DAG).

Phase	Phase Title	Focus Domain	Nodes	Est. Hours
Phase 0	JEE Mathematics Review	Calculus, Matrices, Vectors, Probability, ODEs	8	120 h
Phase 1	Advanced Mathematics	Real Analysis, Linear Algebra, Measure Theory, Convex Optimization	10	250 h
Phase 2	Computational Python	NumPy Vectorization, Pandas, Concurrency, Cython/PyBIND11	6	100 h
Phase 3	Machine Learning	Statistical Learning, Regression, Trees, PyTorch, Model Validation	8	180 h
Phase 4	Time Series Analysis	ARMA/ARIMA, GARCH, Cointegration, Kalman Filters	7	140 h
Phase 5	Financial Economics	MPT, CAPM, Fama-French, Market Microstructure, Fixed Income	6	120 h
Phase 6	Quantitative Finance	Stochastic Calculus, BSM PDE, Volatility Surface, Monte Carlo	8	200 h
Phase 7	Alpha Research	Signal Extraction, Orthogonalization, Backtesting, Risk Models	8	180 h

PHASE	PHASE TITLE	FOCUS DOMAIN	NODES	EST. HOURS
Phase 8	Research Engineering	Modern C++20, Low Latency, Lock-Free SPSC, L3 Order Book	6	160 h
Phase 9	Research Papers	Canonical Paper Replications (Fama, Black-Scholes, Avellaneda, etc.)	10	120 h
Phase 10	Portfolio Projects	Production Alpha Framework, C++ Engine, Volatility Arbitrage	4	200 h
TOTAL CURRICULUM SCOPE			81 Nodes	1,970 Hours

Sample Detailed Node Specification Table

NODE ID	NODE TITLE	HOURS	PREREQUISITES	MASTERY VERIFICATION CRITERIA
P0-1	Limits, Continuity & Differentiation	15 h	None	Solve 20 advanced analytical limits without symbolic solvers.
P1-4	Linear Algebra: Decompositions	30 h	P1-3	Manually derive and compute SVD, Spectral, and QR decompositions.
P4-4	Cointegration & Statistical Arbitrage	25 h	P4-2	Implement Engle-Granger and Johansen cointegration tests for pairs trading.
P6-2	Ito's Lemma & SDEs	30 h	P6-1	Apply Ito's Lemma to derive analytical solutions for Geometric Brownian Motion.
P8-4	Order Book Engine Architecture	30 h	P8-2	Write C++ L3 Order Book maintaining O(1) price-level insertions and cancels.

5. Functional Requirements Specifications

5.1 Dashboard (`FR-DASH`)

Central command center rendering focus items, system stats, revision queues, and progress indicators.

- DASH-W01 Daily Focus Widget:** Displays up to 3 prioritized unlocked topics derived from DAG critical path.
- DASH-W02 Spaced Repetition Widget:** Shows pending SM-2 queue items due today.
- DASH-W03 52-Week Heatmap:** Visualizes daily study duration in monochromatic density shades (`#18181B` to `#A1A1AA`).
- DASH-W04 Velocity Card:** Calculates 14-day rolling average hours/day and pace offset against target completion date.

5.2 Spaced Repetition Engine (`FR-REV`)

Implements a modified SuperMemo-2 (SM-2) algorithm for mathematical concept decay management:

SM-2 MATHEMATICAL FORMULATION

For item i at review iteration n with recall rating $q \in [0, 5]$:

$$EF' = \max\left(1.3, EF + (0.1 - (5 - q) \times (0.08 + (5 - q) \times 0.02)) \times \text{right}\right)$$

$$I_n = 1 \text{ ext{ if } } n=1; \quad 6 \text{ ext{ if } } n=2; \quad I_{n-1} \times EF \text{ ext{ if } } n > 2$$

6. Comprehensive User Story Matrix (100 Specifications)

Below is the complete, non-abbreviated user story execution matrix across all 11 product subsystems.

STORY ID	SUBSYSTEM	USER STORY STATEMENT	ACCEPTANCE TEST CRITERIA
US-DASH-01	Dashboard	As a candidate, I want to view my top 3 daily study priorities upon launch, so that I eliminate decision fatigue.	Displays exactly 3 unlocked nodes sorted by critical path weight.
US-DASH-02	Dashboard	As a candidate, I want to see my total study hours logged today vs my target, so that I stay accountable.	Progress bar renders elapsed hours / target hours in real time.
US-DASH-03	Dashboard	As a candidate, I want to view a 52-week study heatmap, so that I can visualize my study consistency.	52-column grid renders daily log intensity using monochromatic shades.
US-DASH-04	Dashboard	As a candidate, I want to see pending SM-2 revisions due today, so that I can clear reviews immediately.	Counter badge shows items where next_review_date <= today.
US-DASH-05	Dashboard	As a candidate, I want to view my schedule pace offset, so that I know how many days behind or ahead I am.	Displays numerical offset calculated by Velocity Engine.
US-DASH-06	Dashboard	As a candidate, I want to see my overall curriculum completion percentage, so that I track long-term progress.	Renders total completed node hours / 1,970 total hours.
US-DASH-07	Dashboard	As a candidate, I want a single-click button to start the global study timer, so that tracking is frictionless.	Header timer button launches active timing session.
US-DASH-08	Dashboard	As a candidate, I want to view my active Phase progress card, so that I know my current primary focus domain.	Renders current Phase progress bar and completed node ratio.
US-DASH-09	Dashboard	As a candidate, I want to see recently edited notes, so that I can quickly jump back into recent contexts.	Displays 5 most recently updated markdown files.
US-DASH-10	Dashboard	As a candidate, I want to view active project deliverables, so that I can track coding implementation status.	Renders pending check items for active Phase 10 project.
US-PLAN-01	Daily Planner	As a candidate, I want unfinished tasks to carry over automatically to the next day, so no work is dropped.	Uncompleted items retain active status in next day's plan.
US-PLAN-02	Daily Planner	As a candidate, I want to add manual ad-hoc tasks to my daily plan, so that I can handle unexpected study items.	Inline text entry creates new daily task element.

STORY ID	SUBSYSTEM	USER STORY STATEMENT	ACCEPTANCE TEST CRITERIA
US-PLAN-03	Daily Planner	As a candidate, I want to log qualitative reflection notes at the end of the day, so I can track mental state.	Markdown text area saves content to daily_logs table.
US-PLAN-04	Daily Planner	As a candidate, I want to rate my daily focus level from 1 to 5, so I can analyze cognitive performance trends.	5-star selection control records focus score.
US-PLAN-05	Daily Planner	As a candidate, I want to reorder daily tasks via drag-and-drop, so I can customize task execution order.	Reordering persists display_order index in local database.
US-PLAN-06	Daily Planner	As a candidate, I want to attach estimated minutes to manual tasks, so my daily duration is accurately calculated.	Input field updates calculated daily total estimated time.
US-PLAN-07	Daily Planner	As a candidate, I want to mark tasks as completed with a single keypress, so log updates are instant.	Pressing Space or clicking checkbox updates item status.
US-PLAN-08	Daily Planner	As a candidate, I want to view past daily logs by date picker, so I can review historical productivity.	Date picker populates historic logs, ratings, and focus scores.
US-PLAN-09	Daily Planner	As a candidate, I want daily timers to accumulate time directly into active tasks, so manual logging is minimized.	Timer stop action appends recorded duration to target task.
US-PLAN-10	Daily Planner	As a candidate, I want to export daily logs as a Markdown summary, so I can archive my notes externally.	Export action outputs formatted Markdown document.
US-ROAD-01	Roadmap	As a candidate, I want locked nodes to show missing prerequisites, so I don't study out of order.	Hovering locked node displays tooltip listing incomplete parent nodes.
US-ROAD-02	Roadmap	As a candidate, I want completed nodes to automatically unlock dependent child nodes, updating instantly.	Node completion triggers DAG re-evaluation and unlocks eligible nodes.
US-ROAD-03	Roadmap	As a candidate, I want to filter the roadmap by Phase, so I can reduce visual complexity.	Phase tab selector isolates specific phase sub-graphs.
US-ROAD-04	Roadmap	As a candidate, I want the critical path to be highlighted, so I know which sequence dictates end date.	Critical path nodes render with high-contrast accent border.
US-ROAD-05	Roadmap	As a candidate, I want to search for nodes by topic name or keyword, so I can navigate the graph quickly.	Search matches highlight target nodes and pan canvas to focus.
US-ROAD-06	Roadmap	As a candidate, I want to view total estimated hours vs elapsed hours per Phase, so I can monitor pace.	Phase headers render aggregate estimate vs actual duration bar.
US-ROAD-07	Roadmap	As a candidate, I want to pan and zoom smoothly across the graph canvas, so I can inspect dependencies comfortably.	Canvas supports 60fps pan/zoom controls via mouse/wheel.
US-ROAD-08	Roadmap	As a candidate, I want to manually override a node status if I have prior mastery, so I can skip familiar topics.	Status override modal sets node state to MASTERED with confirmation.

STORY ID	SUBSYSTEM	USER STORY STATEMENT	ACCEPTANCE TEST CRITERIA
US-ROAD-09	Roadmap	As a candidate, I want node lines to indicate dependency type, so I can differentiate hard vs soft prerequisites.	Solid vs dashed vectors visually separate core vs optional prerequisites.
US-ROAD-10	Roadmap	As a candidate, I want to view node difficulty badges, so I can gauge topic complexity.	Difficulty tags (Beginner, Advanced, Institutional) rendered on node cards.
US-TOPIC-01	Topic View	As a candidate, I want to write local Markdown notes with LaTeX rendering, so I can document math formulas.	Integrated editor renders block and inline LaTeX seamlessly.
US-TOPIC-02	Topic View	As a candidate, I want to view mastery verification criteria on each topic, so I know when I have achieved true depth.	Mastery section displays explicit checklist required for completion.
US-TOPIC-03	Topic View	As a candidate, I want to link learning resources directly to a topic, so my study materials stay organized.	Resource picker attaches books, videos, and papers to topic record.
US-TOPIC-04	Topic View	As a candidate, I want to view logged study sessions for a specific topic, so I can track effort distribution.	History table lists all recorded time entries for current topic.
US-TOPIC-05	Topic View	As a candidate, I want to create SM-2 flashcard review items directly from topic notes, so I can retain concepts.	Highlighting text allows 'Create Revision Item' context command.
US-TOPIC-06	Topic View	As a candidate, I want to jump directly to parent or child topic views, so navigation is fluid.	Clicking dependency tags opens target topic detail view immediately.
US-TOPIC-07	Topic View	As a candidate, I want topic notes to auto-save instantly, so I never lose typed thoughts.	Debounced auto-save writes changes to SQLite after 500ms idle.
US-TOPIC-08	Topic View	As a candidate, I want to attach practice problems and solutions to a topic, so I can test my understanding.	Practice problem tab allows accordion list of problems and solutions.
US-TOPIC-09	Topic View	As a candidate, I want to view estimated vs actual completion time ratio, so I know if I underestimated.	Ratio badge displays actual_hours / estimated_hours.
US-TOPIC-10	Topic View	As a candidate, I want to export a topic note as a formatted PDF, so I can share or print clean math notes.	Export button generates single-topic standalone PDF document.
US-RESC-01	Resources	As a candidate, I want to categorize resources by medium type, so I can filter books, videos, and papers.	Filter menu isolates Books, PDFs, Videos, Repos, and Documentation.
US-RESC-02	Resources	As a candidate, I want to rate resources from 1 to 5 stars, so I can record resource quality.	Star rating control updates resource database rating.
US-RESC-03	Resources	As a candidate, I want to mark resources as 'In Progress' or 'Completed', so I track reading progress.	Status selector updates resource completion state.
US-RESC-04	Resources	As a candidate, I want to attach local PDF paths to resources, so I can open them in my system viewer.	Local file link executes system default PDF viewer action.
US-RESC-05	Resources	As a candidate, I want to bookmark high-value resources, so I can reference them quickly.	Bookmark toggle adds resource to quick-access sidebar collection.

STORY ID	SUBSYSTEM	USER STORY STATEMENT	ACCEPTANCE TEST CRITERIA
US-RESC-06	Resources	As a candidate, I want to search resources by author or title, so I can locate specific references.	Search input filters resource table dynamically.
US-RESC-07	Resources	As a candidate, I want to view all resources attached to a specific Phase, so I can plan material acquisition.	Phase filter aggregates all child topic resources.
US-RESC-08	Resources	As a candidate, I want to record personal notes on each resource, so I can summarize key takeaways.	Resource detail modal includes dedicated Markdown notes tab.
US-RESC-09	Resources	As a candidate, I want to track page numbers or timestamps for resources, so I resume reading easily.	Progress field stores page X of Y or timestamp MM:SS.
US-RESC-10	Resources	As a candidate, I want to import resource references from BibTeX files, so I can add academic citations.	BibTeX parser ingests entries into resource table.
US-PROJ-01	Projects	As a candidate, I want project deliverable checklists, so I can systematically complete software builds.	Interactive checklist updates total project completion percentage.
US-PROJ-02	Projects	As a candidate, I want to store local GitHub repository paths on projects, so I can open them in IDE.	Repository link button launches terminal or editor at local path.
US-PROJ-03	Projects	As a candidate, I want to link projects to multiple topic nodes, so cross-disciplinary applications are clear.	Multi-select node picker maps projects across relevant curriculum topics.
US-PROJ-04	Projects	As a candidate, I want to track total hours spent coding each project, so I monitor engineering time.	Project-specific timer appends elapsed hours to project record.
US-PROJ-05	Projects	As a candidate, I want to record project architecture decisions in a note tab, so I document engineering design.	Design doc tab stores architecture specs and trade-offs.
US-PROJ-06	Projects	As a candidate, I want to attach performance benchmarks to C++ and Python projects, so I track latency metrics.	Benchmark log table stores operation speeds (e.g., ops/sec, latency ns).
US-PROJ-07	Projects	As a candidate, I want to mark projects as 'Building', 'Testing', or 'Completed', so lifecycle state is clear.	Status selector updates project milestone state.
US-PROJ-08	Projects	As a candidate, I want project milestones to feed into overall Phase completion, so coding counts toward mastery.	Project completion weight contributes to Phase progress calculations.
US-PROJ-09	Projects	As a candidate, I want to log bug notes and edge cases discovered, so I remember complex fixes.	Issue log tab allows recording bugs and resolution notes.
US-PROJ-10	Projects	As a candidate, I want to showcase completed project summaries on a Portfolio View, so I can prepare interview talking points.	Portfolio tab generates unified view of all completed project specs.
US-PAPR-01	Paper Tracker	As a candidate, I want to store key paper equations in LaTeX, so I can reference core proofs.	Paper detail view renders equation blocks with full math syntax.
US-PAPR-02	Paper Tracker	As a candidate, I want to set a 'Revisit Date' for paper reviews, so they integrate into my spaced repetition queue.	Revisit date picker injects paper review into SM-2 schedule.
US-PAPR-03	Paper Tracker	As a candidate, I want to log strategy ideas generated by papers, so I preserve alpha hypotheses.	Ideas log text box stores actionable quantitative hypotheses.

STORY ID	SUBSYSTEM	USER STORY STATEMENT	ACCEPTANCE TEST CRITERIA
US-PAPR-04	Paper Tracker	As a candidate, I want to tag papers by domain (e.g., Microstructure, Volatility), so I can group literature.	Tag input field enables multi-tag filtering across paper library.
US-PAPR-05	Paper Tracker	As a candidate, I want to record paper citation metadata (Authors, Journal, Year), so citations are accurate.	Metadata fields store complete academic reference details.
US-PAPR-06	Paper Tracker	As a candidate, I want to attach code replication repos to paper records, so I track empirical implementations.	Code repository path field links paper directly to replication code.
US-PAPR-07	Paper Tracker	As a candidate, I want to mark paper status as 'Annotated' or 'Replicated', so reading depth is tracked.	Status dropdown tracks paper progression lifecycle.
US-PAPR-08	Paper Tracker	As a candidate, I want to write executive summaries for each paper, so I can recall core takeaways in 30 seconds.	Summary text field renders formatted Markdown overview.
US-PAPR-09	Paper Tracker	As a candidate, I want to link papers to relevant curriculum nodes, so literature relates to underlying math.	Node selector binds paper record to specific topic IDs.
US-PAPR-10	Paper Tracker	As a candidate, I want to export paper reading lists as annotated bibliographies, so I can share literature reviews.	Export command outputs formatted Markdown/LaTeX bibliography.
US-REV-01	Revision Queue	As a candidate, I want recall quality scores (0-5) to dynamically update SM-2 intervals, so hard items repeat sooner.	Score submission recalculates Ease Factor and next review date.
US-REV-02	Revision Queue	As a candidate, I want to view pending revision counts grouped by topic, so I see concept decay focus areas.	Queue breakdown displays pending counts per topic node.
US-REV-03	Revision Queue	As a candidate, I want forgotten items ($q < 3$) to reset review interval to 1 day, so I restudy lost concepts immediately.	Sub-3 recall score resets interval to 1 and increments lapse count.
US-REV-04	Revision Queue	As a candidate, I want to flip flashcards to reveal LaTeX math solutions, so I test my active recall.	Interactive card component toggles question and answer views.
US-REV-05	Revision Queue	As a candidate, I want to view historical review accuracy stats, so I measure memory retention performance.	Analytics chart plots retention rate percentage over time.
US-REV-06	Revision Queue	As a candidate, I want to launch a quick 10-minute revision session, so I can review concepts in short breaks.	Quick Session mode caps review queue to top 10 highest-decay items.
US-REV-07	Revision Queue	As a candidate, I want to suspend specific flashcards if they are obsolete, so my queue stays clean.	Suspend action toggles card active state without deleting historical data.
US-REV-08	Revision Queue	As a candidate, I want to tag flashcards by formula type, so I can run targeted drills on specific sub-topics.	Tag selector filters active review session to matching tags.

STORY ID	SUBSYSTEM	USER STORY STATEMENT	ACCEPTANCE TEST CRITERIA
US-REV-09	Revision Queue	As a candidate, I want to view card difficulty distribution, so I know which topics have lowest Ease Factors.	Chart renders distribution of cards across Ease Factor bands.
US-REV-10	Revision Queue	As a candidate, I want card creation to automatically embed surrounding topic context, so cards retain source links.	Cards store source topic ID and render clickable origin breadcrumb.
US-ANLY-01	Analytics	As a candidate, I want to view my forecasted completion date based on 14-day velocity, so I manage my timeline.	Engine computes target completion date using rolling average velocity.
US-ANLY-02	Analytics	As a candidate, I want to view weekly study hour reports, so I can evaluate weekly throughput consistency.	Bar chart renders aggregated study hours per week for last 16 weeks.
US-ANLY-03	Analytics	As a candidate, I want to see a breakdown of time spent by Phase, so I identify where most effort is consumed.	Donut chart displays percentage distribution of hours across all 11 phases.
US-ANLY-04	Analytics	As a candidate, I want to track my longest daily study streak, so I stay motivated to maintain momentum.	Streak counter tracks consecutive days with logged hours $\geq$ minimum target.
US-ANLY-05	Analytics	As a candidate, I want to compare actual study time against estimated topic hours, so I refine future estimates.	Scatter plot correlates estimated vs actual hours across completed nodes.
US-ANLY-06	Analytics	As a candidate, I want to view learning velocity trends over time, so I see if my study capacity is increasing.	Line graph plots rolling 14-day velocity over 6-month window.
US-ANLY-07	Analytics	As a candidate, I want to measure resource completion rates, so I know how much of my reading list is finished.	Progress bar renders completed resources / total resources.
US-ANLY-08	Analytics	As a candidate, I want to view monthly productivity summaries, so I can perform high-level monthly reviews.	Monthly report card aggregates total hours, nodes completed, and papers read.
US-ANLY-09	Analytics	As a candidate, I want to export raw study session logs as CSV, so I can run custom data analyses in Python.	CSV export dumps complete session log dataset to local file.
US-ANLY-10	Analytics	As a candidate, I want to track average recall score accuracy over time, so I verify memory stability.	Trend chart plots monthly mean recall score from SM-2 reviews.
US-SYS-01	System	As a candidate, I want all app data stored in a single local SQLite database, so I own my data entirely.	Application initializes and persists state exclusively to local SQLite file.
US-SYS-02	System	As a candidate, I want to export my entire database to JSON, so I can perform complete offline backups.	Export command writes full database tables into single valid JSON file.
US-SYS-03	System	As a candidate, I want to import a JSON backup file, so I can restore state across desktop devices.	Import parser validates schema and hydrates database tables atomically.
US-SYS-04	System	As a candidate, I want the app to run completely disconnected from the internet, so I study in isolated environments.	Zero outbound network connections confirmed during application runtime execution.

STORY ID	SUBSYSTEM	USER STORY STATEMENT	ACCEPTANCE TEST CRITERIA
US-SYS-05	System	As a candidate, I want automatic database snapshot backups on launch, so I am protected against disk corruption.	App creates rolling snapshot copies of SQLite file in local backup folder.
US-SYS-06	System	As a candidate, I want to configure my daily study target hours in settings, so progress bars adapt to my schedule.	Setting field updates baseline target hours used across dashboard gauges.
US-SYS-07	System	As a candidate, I want to export all topic notes as an Obsidian-compatible folder tree, so I retain open Markdown files.	Export script outputs nested folder hierarchy containing formatted .md files.
US-SYS-08	System	As a candidate, I want customizable light and dark themes with high-contrast, distraction-free styling.	Theme toggle updates global CSS variables across all interface components.
US-SYS-09	System	As a candidate, I want system notifications when study timers complete, so I take scheduled breaks.	Native desktop alert fires upon timer completion.
US-SYS-10	System	As a candidate, I want lightweight memory usage (< 150MB RAM), so the app runs smoothly alongside code IDEs.	Process monitor confirms RAM footprint remains under specified threshold.
US-CMD-01	Command Palette	As a candidate, I want to trigger a universal command modal using Cmd+K, so I navigate without mouse movement.	Global hotkey Cmd+K / Ctrl+K opens focused overlay across all views.
US-CMD-02	Command Palette	As a candidate, I want fuzzy search across all nodes, papers, and projects, so I open items in under 2 seconds.	Fuzzy search engine returns filtered results within < 15ms latency.
US-CMD-03	Command Palette	As a candidate, I want to execute instant actions like 'Start Timer' directly from Cmd+K, so workflow is rapid.	Typing action command triggers corresponding function instantly.
US-CMD-04	Command Palette	As a candidate, I want recent searches to appear at the top of Cmd+K modal, so I reopen frequent views quickly.	Modal lists last 5 executed commands / navigation targets on open.
US-CMD-05	Command Palette	As a candidate, I want full keyboard arrow key navigation in Cmd+K, so I select items seamlessly.	Up/Down arrows cycle selection; Enter executes target item.
US-CMD-06	Command Palette	As a candidate, I want to search for specific LaTeX equations in paper notes, so I find math formulas instantly.	Search engine indexes math blocks inside paper notes.
US-CMD-07	Command Palette	As a candidate, I want keyboard shortcuts displayed next to menu items, so I learn direct hotkeys over time.	Cmd+K list displays registered shortcut badges alongside actions.
US-CMD-08	Command Palette	As a candidate, I want Esc to instantly close the command modal, returning focus to main interface.	Pressing Esc closes overlay and restores previous focus target.
US-CMD-09	Command Palette	As a candidate, I want to jump directly to specific daily log dates from Cmd+K, so I inspect history quickly.	Typing 'log [date]' opens corresponding daily reflection view.

STORY ID	SUBSYSTEM	USER STORY STATEMENT	ACCEPTANCE TEST CRITERIA
US-CMD-10	Command Palette	As a candidate, I want search results grouped by category (Nodes, Papers, Projects, Commands), so options are clear.	Results list renders categorized header dividers above matching entries.

7. Conceptual Data Architecture & Local Schema

All data is persisted locally in an ACID-compliant SQLite database. Below are core SQL schema definitions representing the primary entities.

```
-- Core Curriculum Node Table
CREATE TABLE topic_nodes (
  id TEXT PRIMARY KEY,          -- e.g. 'P6-4'
  title TEXT NOT NULL,
  phase_id TEXT NOT NULL,       -- e.g. 'PHASE_6'
  estimated_hours REAL NOT NULL,
  status TEXT NOT NULL DEFAULT 'LOCKED', -- LOCKED, UNLOCKED, IN_PROGRESS, COMPLETED, MASTERED
  description TEXT,
  mastery_criteria TEXT,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- Node Dependency Relations (DAG Edges)
CREATE TABLE topic_dependencies (
  topic_id TEXT NOT NULL,
  prerequisite_topic_id TEXT NOT NULL,
  PRIMARY KEY (topic_id, prerequisite_topic_id),
  FOREIGN KEY (topic_id) REFERENCES topic_nodes(id),
  FOREIGN KEY (prerequisite_topic_id) REFERENCES topic_nodes(id)
);

-- Spaced Repetition Items (SM-2 Engine)
CREATE TABLE spaced_repetition_items (
  id TEXT PRIMARY KEY,
  topic_id TEXT NOT NULL,
  prompt TEXT NOT NULL,
  answer TEXT NOT NULL,
  ease_factor REAL DEFAULT 2.5,
  interval_days INTEGER DEFAULT 1,
  review_count INTEGER DEFAULT 0,
  next_review_date DATE NOT NULL,
  FOREIGN KEY (topic_id) REFERENCES topic_nodes(id)
);

-- Precision Study Session Logs
CREATE TABLE study_sessions (
  id TEXT PRIMARY KEY,
  topic_id TEXT NOT NULL,
  duration_hours REAL NOT NULL,
  session_timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
  notes TEXT,
  FOREIGN KEY (topic_id) REFERENCES topic_nodes(id)
);
```

8. UI/UX Specifications & Visual System

QuantOS adopts a strict, distraction-free aesthetic inspired by high-density developer tools (Linear, Raycast, Obsidian, Apple Notes). No decorative ornaments, Greek statues, marble textures, or Latin terminology are permitted.

- **Color System (Monochromatic Dark):** Canvas ``#0D0E12``, Panels ``#121318``, Borders ``#27272A``, Muted Text ``#A1A1AA``, Contrast Text ``#FAFAFA``, Primary Accent ``#38BDF8`` (Muted Ice Blue).
- **Typography:** UI Text in ``Inter`` / ``-apple-system``; Code, Math, and Metrics in ``JetBrains Mono``.
- **Density & Space:** Compact 4px grid spacing, high information density, clear visual hierarchy, keyboard-first controls.

9. Non-Functional Requirements (NFRs)

NFR ID	CATEGORY	REQUIREMENT METRIC	TARGET THRESHOLD
<code>NFR-PERF-01</code>	Performance	Cold Application Launch Latency	< 300 ms on baseline hardware
<code>NFR-PERF-02</code>	Performance	UI Frame Rate during Pan/Zoom	Consistent 60 FPS (< 16.6 ms per frame)
<code>NFR-PERF-03</code>	Performance	Command Palette Search Latency	< 15 ms across 10,000 indexed records
<code>NFR-REL-01</code>	Reliability	Auto-Save Interval	State saved after 500 ms idle typing
<code>NFR-REL-02</code>	Integrity	Database Transaction Safety	100% ACID-compliant SQLite operations
<code>NFR-ACC-01</code>	Accessibility	Keyboard Navigation	100% functional coverage without mouse mouse input

10. Acceptance Criteria Matrix

FEATURE MODULE	REQ ID	VERIFICATION TEST SCENARIO	EXPECTED PASS RESULT
DAG Roadmap	<code>FR-ROAD-01</code>	Mark all parent nodes of <code>`P1-4`</code> as COMPLETED.	<code>`P1-4`</code> status changes instantly from LOCKED to UNLOCKED.
SM-2 Revision Engine	<code>FR-REV-01</code>	Submit recall quality <code>`q = 5`</code> for card item.	Ease factor increases, interval multiplies, <code>next_review_date</code> advances.
Daily Planner	<code>FR-PLAN-01</code>	Leave task uncompleted on Day N. Advance clock to Day N+1.	Task automatically carries forward to Day N+1 active queue.
Data Portability	<code>FR-SYS-01</code>	Execute <code>`Cmd+K`</code> -> Export Full Database.	Outputs valid single-file JSON dump containing all schema tables.

11. Risk Analysis & Mitigation Strategy

RISK ID	RISK DESCRIPTION	IMPACT	LIKELIHOOD	MITIGATION STRATEGY
<code>R-01</code>	Curriculum Overwhelm causing early abandonment.	High	Medium	Enforce Daily Focus Engine hiding macro DAG complexity behind 3 daily tasks.
<code>R-02</code>	Time Estimation Variance distorting velocity forecasts.	Medium	High	Adaptive recalibration: engine automatically adjusts downstream estimates based on past pace.
<code>R-03</code>	Local Disk Corruption causing progress data loss.	High	Low	Automated launch snapshot backups keeping rolling 10 database versions.

12. Future Product Roadmap (V2 & V3)

Version 2.0: LaTeX Monograph Publisher • Jupyter Notebook Local Sandbox • VS Code & CLion Deep-Link Plugins.

Version 3.0: Peer-to-Peer Local Mesh Graph Sync • Integrated Lightweight C++ Backtesting Simulator • Automated Interview Question Generator.

13. Appendices & Glossary

QuantOS: Quantitative Operating System; the offline-first desktop environment defined in this specification.

DAG: Directed Acyclic Graph; a mathematical graph of nodes connected by directed edges containing no closed loops.

SM-2: SuperMemo-2 algorithm; a mathematical model calculating optimal interval expansion for human memory retention.

Alpha Signal: A predictive mathematical formula or feature forecasting risk-adjusted asset returns.